University of Nottingham
UK | CHINA | MALAYSIA

EEEE4064 Advanced Control System Design - Coursework 1 Marking Rubric

Component	Poor	Pass	Good	Excellent
System identification (20%)	**0–7**	**8–11**	**12–17**	**18–20**
	No defensible identified model, or experimental responses/input are interpreted incorrectly. Major identification or physical inconsistencies; fit evidence is missing, unreliable, or well below the stated requirement.	A model is obtained, but data/input handling, fit, structure, or model interpretation has significant weaknesses. Meets only some expectations and provides little convincing evidence for the chosen model.	Experimental data and input are handled correctly. A suitable identified model meets the published accuracy requirement and captures the main measured dynamics. Model complexity and adequacy are mostly defensible, with minor limitations.	The identified model is accurate, parsimonious, physically credible, and supported by convincing quantitative verification. Structure and assumptions are justified; important discrepancies and limitations are critically examined. Full marks require exceptional adequacy with no material unresolved issue.
System modeling (40%)	**0–15**	**16–23**	**24–35**	**36–40**
	The problem is not formulated coherently. Mathematical models or state-space representations are absent, dimensionally inconsistent, or fundamentally incorrect; little credible first-principles reasoning is shown.	A partially valid mathematical formulation and model are presented, but there are significant gaps in derivation, variables, assumptions, model relationships, or matrix consistency. Verification is limited or unreliable.	Correct and coherent mathematical models and compatible representations are developed. Formula-based derivations are largely rigorous, dimensions and assumptions are generally appropriate, and key results are verified independently using MATLAB; minor gaps remain.	A comprehensive, mathematically rigorous and traceable model is developed from correctly formulated system information. Relevant state-space and equivalent representations, transformations, and derivations are internally consistent, justified, and independently verified. Full marks require no substantive algebraic, structural, or equivalence error.
System analysis (30%)	**0–11**	**12–17**	**18–26**	**27–30**
	System properties are largely unexamined or assessed using invalid criteria. Major errors or unsupported claims affect stability, controllability, observability, or response interpretation.	Some relevant system properties are investigated, but significant results, analytical checks, or justified conclusions are missing or incorrect. Numerical outputs are often reported without explaining their engineering meaning.	System dynamics and key properties are analyzed correctly using suitable mathematical criteria. Analytical and MATLAB results are compared, response characteristics are evaluated appropriately, and conclusions are largely substantiated; minor gaps remain.	A rigorous, critical, and well-integrated analysis explains the model's dynamic behavior, stability, controllability, observability, and relevant transient characteristics. Independent checks, numerical precision, assumptions, and limitations are addressed. Full marks require accurate, comprehensive evidence and technically defensible conclusions without material omission.
Code quality (10%)	**0–3**	**4–5**	**6–8**	**9–10**
	The script fails substantially or depends on undocumented workspace variables/files. Calculations and results are difficult to locate, reproduce, or interpret; organization and comments are inadequate.	Core code is partly functional but needs significant manual intervention or correction. Sections, comments, outputs, units, or dependencies are inconsistent; reproducibility and traceability are limited.	The script is organized, readable, and largely executable from the specified files. Comments are concise and useful, outputs/figures are sufficiently identified, and calculations are traceable; only minor coding or reproducibility weaknesses remain.	The script runs reliably in a clean MATLAB environment with the specified inputs, produces clear and traceable evidence, and uses precise organization, units, labels, and concise technical comments. Full marks require complete reproducibility, robust handling of foreseeable issues, and no hidden dependencies or avoidable coding defects.

Below acceptable outcome attainment.	Basic, incomplete attainment.	Substantial, competent attainment.	Exceptional, fully substantiated attainment.